AMOXCALLI · VERSION 0.1.0 · OCTOBER 2026

Learning to read Classical Nahuatl with evidence

A local, open tool: every answer says where it comes from —a dictionary entry, a passage, a pictographic record or a rule. **Without evidence, it abstains.**

12
guided readings
48 passages from 10 works

16,111
bilingual passages from the Axolotl archive

3,790
sourced dictionary senses

0
LLMs required: runs on standard Python



THE PROBLEM

Reading a classical text today takes three things that are not in one place

Scattered sources

Colonial dictionaries, codices, glyph catalogues and parallel corpora live on different sites, under different licences. A beginner cannot tell how to get back to the document.

Many spellings

The same word is written *qualli*, *cualli*, *kwalli* or *kuali* depending on the source, the convention or the variety. An exact search for *cualli* finds 293 of its 486 occurrences.

Tools that make things up

A machine translator produces plausible text with no source. We tested one: our experimental translator (Qwen3-0.6B + LoRA) scored **below** simply returning the most similar passage (22.4 vs 27.2 chrF++). It was retired.

"Good" in Amoxcalli's archive: *cualli* 293 *kuali* 74 *cuali* 68 *qualli* 42 *cualii* 3 *kwali* 3 *quali* 2 *kwalli* 1

Amoxcalli's bet: bring reading, analysis, spellings, dictionary, glyphs and archive together, and generate nothing that a visible source cannot back up.

Source: README and reports/EVALUACION.md

DEMO 1 · A GUIDED READING

Tap a word and see how it is built

Lecturas → Reconocer una flor (“Recognising a flower”). A passage from *La tinta negra y roja*: *NIQUITTA IN XOCHITL*.

- *NIQUITTA* → *ni-qu-itta* · “I-it-see”, with an interlinear gloss.
- The help says what the source record translates (“contemplo”, I contemplate) and that the analysis of this form is still pending.
- The related glyph is presented as a **relation of meaning**, not as the way the word is written.
- Hideable translation, exercises on the passage’s own spelling, and comprehension questions quoted from the original translation.

12 readings · 48 real passages from 10 works · every word can be analysed.



Screenshot of the app. Glyph images load from their own sites (here, a box with the credit): the project does not copy them.

The screen itself says “automatic interlinear gloss (rules + dictionary + corpus)”.

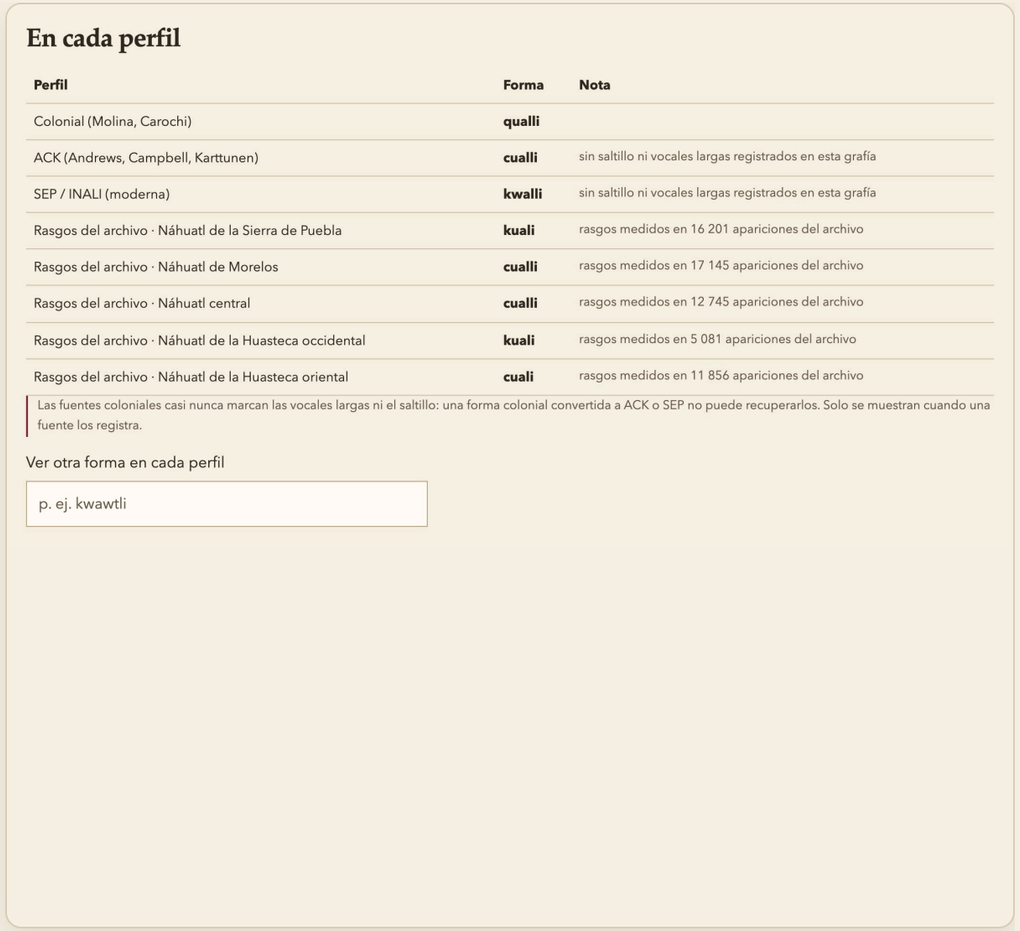
DEMO 2 · SPELLING SEARCH

Type it the way you write it: *kuali* leads to *cualli*, with sources

- "*kuali*" and "*cualli*" are the same form in another convention · **high confidence** · 486 occurrences.
- Every candidate says **why** it is proposed and how sure it is; a guess is never shown as an attested form.
- **It relates spellings; it does not impose a standard.** No spelling is "the correct one".



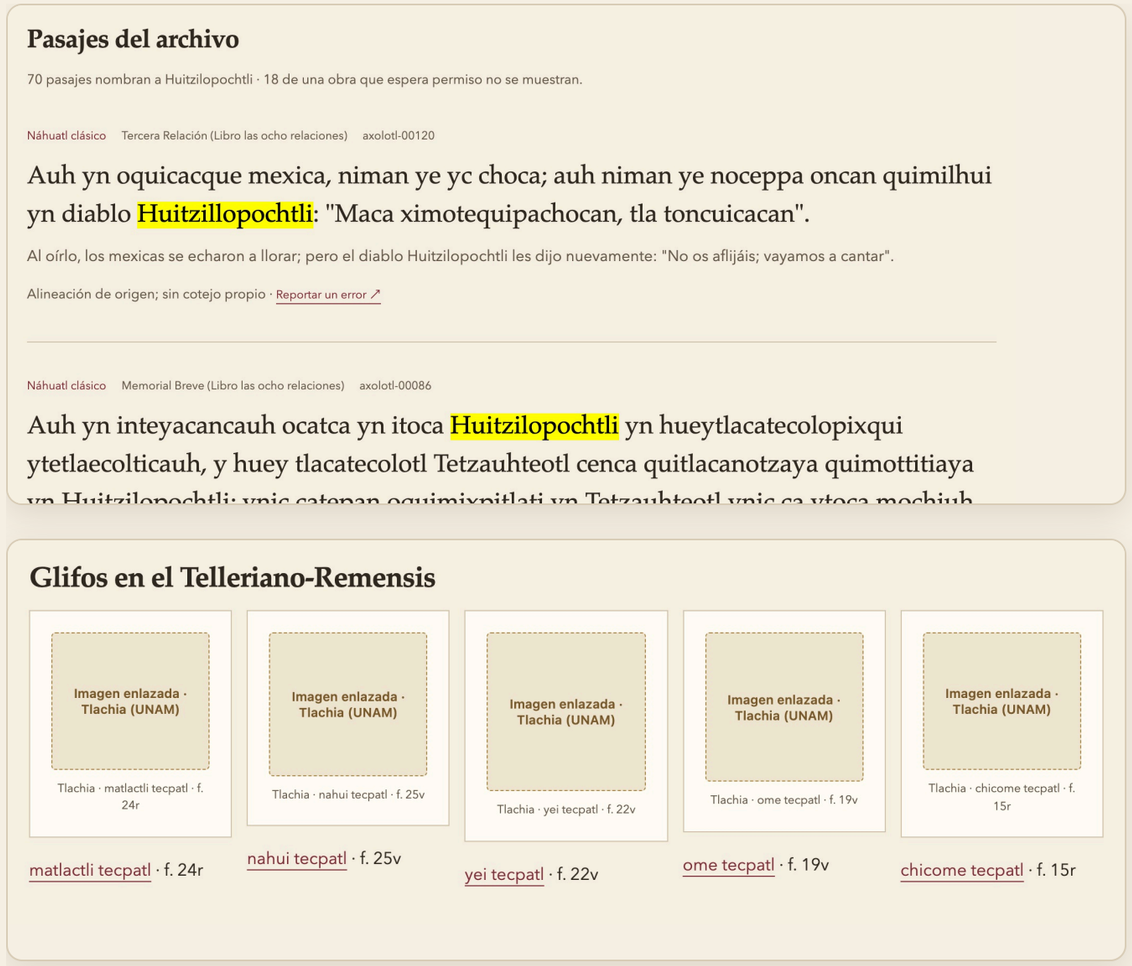
Typing *kuali* in the search box.



The same word in the colonial (*qualli*), ACK (*cualli*) and SEP/INALI (*kwalli*) profiles, and in the traits measured for each variety. The screen warns that colonial sources almost never mark vowel length or the glottal stop.

DEMO 3 · DEITIES AND CALENDAR

Huitzilopochtli: spellings, passages and a place in the year



The *tecpatl* sign: its glyphs in the Codex Telleriano-Remensis, with folio. Tlachia (UNAM) images are linked; here, a box with the credit.

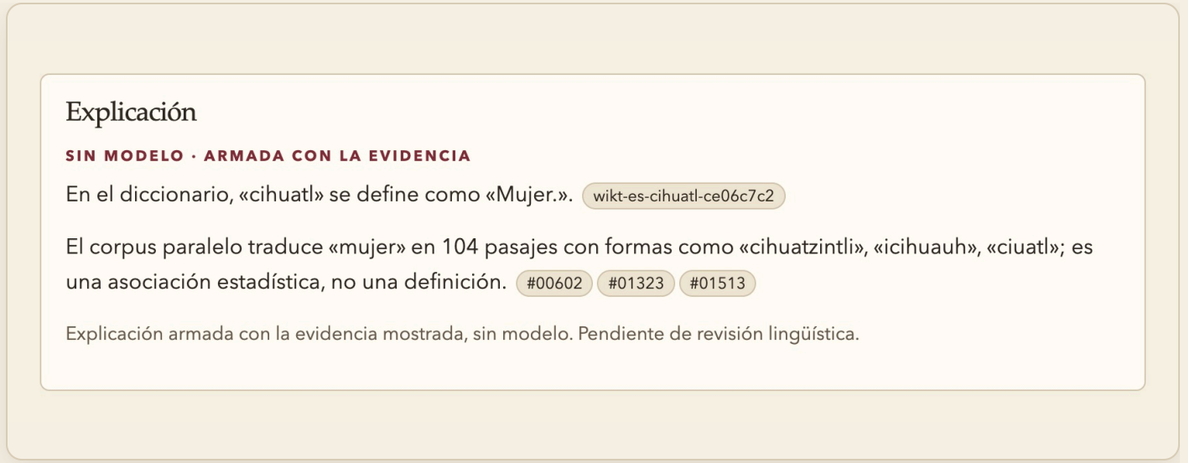
- Searching "*Huitzilopoxtli*", Ask links the card: *possible spelling via x = ch*, medium confidence.
- **8 spellings** from the archive (*huitzillopochtli, uitzilopochtli, vitzilopuchtli…*) lead to the same card.
- **70 passages** name him; 18 from a work awaiting permission **are counted but not shown**.
- Every statement cites its source; the card separates Chimalpahin's voice ("devil") from the older tradition.
- In the calendar: the veintena dedicated to him, *panquetzaliztli* (source: Wiktionary).
- **Glyphs:** the bundled Tlachia catalogue has no record for Huitzilopochtli, and the card does not invent one. The 20 day signs of the tonalpohualli do carry their Telleriano-Remensis glyph.

Borrador · sin revisar (draft · unreviewed) 26 deities · 20 signs · 20 trecenas · 18 veintenas

HOW EACH ANSWER IS BACKED

Cited evidence, abstention, and automatic content in plain view

- **Evidence:** in Ask (Preguntar), every sentence of the explanation cites the IDs of what is shown. A validator rejects uncited sentences, IDs outside the evidence and quoted forms missing from the cited text.
- **Abstention:** with no entry or passage to support it, there is no answer: “no translation is proposed without evidence”.
- **Automatic content is labelled:** “automatic analysis · reading hypothesis”, “draft · unreviewed”, “automatic extraction”. “Reviewed by…” appears only with a decision signed by a person.
- **No LLM:** everything works without a model. Explanations written by Claude are optional, switched on by hand, and shown only if they pass the same validator.
- **No data leaves the computer** except with that option; no account, no telemetry. (Glyph images load from their own sites.)



“¿Cómo se dice mujer?” (how do you say woman?): two sentences, four citations, no model.



“¿Cómo se dice computadora?” (computer): not enough evidence.

HONEST NUMBERS

What we measured, and what these numbers do not say

MEASUREMENT (TEST SPLIT WHERE THE SET HAS ONE)	RESULT	READING
Morphological segmentation, blind test (91 forms written before measuring)	78%	words 67%, particles 94%, proper names 100%
Classical corpus tokens with a root linked to the dictionary	60.4%	41.8% before the Wiktionary batch
Spelling search: intended word first / in the top three	94.2 / 97.1%	0 wrong merges; covers 81.5% of pairs
Ask: correct entry first	92.3%	vs 42.3% for exact search; always with a relevant passage
Explanations with valid citations · documentary records with exact fields (development)	46/46 · 780/790	citations checked by the validator
Statistical gloss hints matching Molina (precision@1)	30%	hence shown as an association, never as a definition
LoRA translator (historical, chrF++)	22.4	vs 27.2 for the closest passage: not a translator; retired

Limits: small sets written by the technical team. They measure technical behaviour; **they do not replace review by an expert, nor show that anyone learns.** `make eval` regenerates every number; `make check` fails if one changed without being regenerated.

Source: reports/EVALUACION.md

WHAT IS MISSING

Three things software cannot do on its own

Linguistic review

- No reviewer has signed off on anything yet.
- In the queue: 4 generated units and 6 readings, 4 mythology readings and a calendar unit.
- 3,215 of the 3,790 senses are unreviewed Wiktionary extractions (1,184 glossed in English).

Checking primary sources

- Passages say “source alignment; not checked by us”.
- Locators still lack the page of the primary edition.
- The extended GDN and the Florentine Codex await permission: the letters are drafted, not sent.

A trial with beginners

- Pending since September: it needs a partner.
- Proposal: a facilitated workshop with 6 to 8 adults, 20–30 minutes, with a rubric agreed with the teacher.
- Meant to find difficulties, not to generalise.

Also: the analyzer covers Classical Nahuatl only; variety rules (such as a = o in the Sierra de Puebla) await review; *teotleco* and *nemontemi* have no open source for their names; glyph images need a connection.

Sources: docs/ROADMAP.md · docs/PILOTO_INSTITUCIONAL.md

WHAT WE ASK FOR

A small collaboration, with deliverables you can check

Philologists and linguists of Classical Nahuatl

Review one four-passage reading, or 50 dictionary entries.

Every decision is recorded with its source; your name appears as reviewer only if you authorise it.

Institutions that hold sources

Permission to link, quote and incorporate with attribution.

UNAM (Gran Diccionario Náhuatl and Tlachia); the *Digital Florentine Codex*. No model training, no commercial use, removal on request.

Teachers and speaker communities

See your way of writing recognised, and try a facilitated workshop.

Community, variety, goals, recognition and permissions are agreed first. Contributions can be withdrawn.

In return: visible attribution in the form each party chooses · a provenance record with date and hash per entry · removal of any content on request · the tools and reviewed materials, open to collaborators.

> We are not asking for endorsement, logos or adoption. We ask for **one concrete review** that turns an automatic draft into reviewed material, and an honest test of whether it helps.

Full proposal (in Spanish): docs/PILOTO_INSTITUCIONAL.md

HOW TO TRY IT

On your own computer, no internet or account needed

1. Install **Python 3.11** or newer.
2. Download **amoxcalli-v0.1.0.zip** from github.com/pescamill/amoxcalli/releases and unzip it.
3. Double-click **Abrir Amoxcalli.command** (macOS) or **Abrir Amoxcalli.bat** (Windows). It opens in your browser; the first time it prepares the analysis for about a minute.

Five-minute tour: Lecturas → Reconocer una flor → *NIQUITTA* and the gloss · type *kuali* in the search box · ask "¿cómo se dice mujer?" · Más → Dioses y calendario → *Huitzilopoxtli*.

Code and documentation: MIT. Third-party texts, images and data keep their own terms. Passages, entries and cards carry a "Reportar un error" (report an error) link.



Contact: [name] · [email] · issues at github.com/pescamill/amoxcalli